COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE010 | **SECTION:** | CPE21S4 |
| **DATE PERFORMED:** | 7/31/26 | **DATE SUBMITTED:** | 7/31/26 |
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## ACTIVITY NO. 4.1 [Stacks]

### 1. PROCEDURE OUTPUT

What procedures did you perform?

PROCEDURE

ILO A: Create a stack using the C++ STL

Definition from the official CPP documentation for the stack STL.

Stacks are a type of container adaptor, specifically designed to operate in a LIFO context (last-in first-out), where elements are inserted and extracted only from one end of the container.

stacks are implemented as container adaptors, which are classes that use an encapsulated object of a specific container class as its underlying container, providing a specific set of member functions to access its elements. Elements are pushed/popped from the "back" of the specific container, which is known as the top of the stack.

The underlying container may be any of the standard container class templates or some other specifically designed container class. The container shall support the following operations:

SUPPLEMENTARY ACTIVITY

ILO C: Solve problems using an implementation of stack:

The following problem definition and algorithm is provided for checking the balancing of symbols. Create an implementation using stacks. Your output must include the following:

a. Stack using Arrays

b. Stack using Linked Lists

c. (Optional) Stack using C++ STL

Problem Definition:

Stacks can be used to check whether the given expression has balanced symbols. This algorithm is very useful in compilers. Each time the parser reads one character at a time. If the character is an opening delimiter such as (, {, or [- then it is written to the stack. When a closing delimiter is encountered like ), }, or ]-the stack is popped. The opening and closing delimiters are then compared. If they match, the parsing of the string continues. If they do not match, the parser indicates that there is an error on the line.

Steps:

1. Create a Stack.

2. While(end of input is not reached) {

a) If the character read is not a symbol to be balanced, ignore it.

b) If the character is an opening symbol, push it onto the stack.

c) If it is a closing symbol:

i) Report an error if the stack is empty.

ii) Otherwise, pop the stack.

d) If the symbol popped is not the corresponding opening symbol, report an error.

3. At the end of input, if the stack is not empty: report an error.

Self-Checking:

For the following cases, complete the table using the code you created.
Expression Valid? (Y/N) Output (Console Screenshot)
Analysis

(A+B)+(C-D)
((A+B)+(C-D)
((A+B)+[C-D])
((A+B]+[C-D]}

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

B1.

```
#include <iostream>
using namespace std;

const int maxCap = 100;
int stack[maxCap];
int top = -1;
int maxSize;
int newData;

void push();
void pop();
void Top();
bool isEmpty();

int main() {
    int choice;

    cout << "Enter number of maximum elements for the stack (1-100): ";
    cin >> maxSize;

    if (maxSize < 1 || maxSize > maxCap) {
        cout << "Invalid stack size!" << endl;
        return 0;
    }

    do {
        cout << "\n===== STACK OPERATIONS =====" << endl;
```

```
        cout << "1. PUSH" << endl;
        cout << "2. POP" << endl;
        cout << "3. TOP" << endl;
        cout << "4. isEMPTY" << endl;
        cout << "5. EXIT" << endl;
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                push();
                break;

            case 2:
                pop();
                break;

            case 3:
                Top();
                break;

            case 4:
                if (isEmpty())
                    cout << "True - Stack is Empty." << endl;
                else
                    cout << "False - Stack is NOT Empty." << endl;
                break;

            case 5:
                cout << "Exiting program..." << endl;
                break;

            default:
                cout << "Invalid Choice." << endl;
        }

    } while (choice != 5);
```

```
    return 0;
}

bool isEmpty() {
    return (top == -1);
}

void push() {
    if (top == maxSize - 1) {
        cout << "Stack Overflow." << endl;
        return;
    }

    cout << "Enter new value: ";
    cin >> newData;

    stack[++top] = newData;

    cout << newData << " pushed into the stack." << endl;
}

void pop() {
    if (isEmpty()) {
        cout << "Stack Underflow." << endl;
        return;
    }

    cout << "Popped: " << stack[top] << endl;
    top--;
}

void Top() {
    if (isEmpty()) {
        cout << "Stack is Empty." << endl;
        return;
    }

    cout << "Top element: " << stack[top] << endl;
```

```
}
```

B2.

```
#include <iostream>
using namespace std;

class Node {
public:
    int data;
    Node* next;
};

Node *head = NULL, *tail = NULL;

// Push element onto the stack
void push(int newData) {
    Node *newNode = new Node;
    newNode->data = newData;
    newNode->next = head;

    if (head == NULL) {
        head = tail = newNode;
    } else {
        head = newNode;
    }
}

// Pop element from the stack
int pop() {
    if (head == NULL) {
        cout << "Stack Underflow." << endl;
        return -1;
    }

    Node *temp = head;
    int tempVal = temp->data;
```

```
    head = head->next;

    // If stack becomes empty
    if (head == NULL) {
        tail = NULL;
    }

    delete temp;
    return tempVal;
}

// Display top element
void Top() {
    if (head == NULL) {
        cout << "Stack is Empty." << endl;
    } else {
        cout << head->data << endl;
    }
}

int main() {
    push(1);
    cout << "After the first PUSH, top of stack is: ";
    Top();

    push(5);
    cout << "After the second PUSH, top of stack is: ";
    Top();

    cout << "Popped: " << pop() << endl;
    cout << "After the first POP, top of stack is: ";
    Top();

    cout << "Popped: " << pop() << endl;
    cout << "After the second POP, top of stack is: ";
    Top();

    cout << "Popped: " << pop() << endl;
```

```
    return 0;
}
```

| Expression Valid? | (Y/N) | Output (Console Screenshot) | Analysis |
|---|---|---|---|
| (A+B)+(C-D) | Y | | All parentheses are correctly matched. |
| ((A+B)+(C-D) | N | | One opening has no matching |
| ((A+B)+[C-D]) | Y | | Parentheses and brackets are correctly matched. |
| ((A+B]+[C-D]} | N | | Closing symbols do not match their corresponding opening symbols. |

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

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**Extent of Use:**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

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